

become increasingly operational. It is slowly becoming effective in mass-market applications.

Consider a wireless switch, which makes use of the energy-harvesting principle, using piezo-electric energy generated by the switching relay to send a radio frequency (RF) signal to the light source to turn it on or off.

Another example is pressure and wear sensors that mainly work on energy-harvesting technology. Energy harvesting is critically required in wireless sensor networks, and in more complex and power-hungry electronic devices.

Primary components of energy harvesting include the following:

- Transducer
- Power convertor
- Sensor
- Computing, memory and communication circuitry
- Some form of energy storage

Energy-harvesting-based wireless sensors are feasible today because of the availability of new-generation, ultra-low-power microcontrollers (MCUs) that can run control algorithms and transmit data using sophisticated power management techniques.

Gecko EFM 32 MCU from Energy Micro, for example, is a 32-bit MCU based on ARM Cortex 3 Core. It is designed for low-power operations.

Cymbet has teamed up with Texas Instruments for providing support for TIS MSP430 value line launch pad development kit. MSP430 is Texas Instruments' ultra-lower sixteen-bit MCU family designed especially for energy-harvesting applications in measurement, sensing and metering sectors. It features a low supply voltage range, from 1.8V to 3.6V, and low power consumption of 270 micro-ampere in active mode at 1MHz, 2.2V, 0.3 micro-ampere in standby mode and 0.1 micro-ampere in off mode.

Working of the SOTB process

The rapid growth of the IoT requires all applications to be connected wirelessly, which demands a strong need for battery-free operations. Here, products use natural energy sources such as light, vibration and heat, or products that consume less power, to extend battery life. Once a much

longer battery life is achieved, battery replacement will no longer be required, enabling maintenance-free applications.

The SOTB process overcomes several drawbacks of earlier processes. In earlier process technologies, an oxide film (buried oxide) is buried under a thin silicon layer on wafer substrate. SOTB technology adopts dopant-less channel transistors that do not require doping the thin film silicon layer. By making the structure dopant-less, transistor threshold characteristics variations can be reduced to approximately one-third of the earlier structure.

In SOTB process, substrate bias and signal response to computational load are dynamically controlled. This results in high-speed operation and lower power consumption. Embedding SRAM with SOTB structure results in harvesting energy and presents maintenance-free IoT applications.

Renesas FD-SOI is SOTB-based, and is used to charge lithium secondary batteries used in healthcare, wearables and hearing aids.

Connecting MCUs for harvesting energy

Renesas is the first company to develop a commercial MCU using SOTB technology in the form of R7FOE embedded controller. It is a 32-bit Arm Cortex-based embedded controller capable of operating up to 64MHz. This MCU can rapidly process local sensor data and execute complex analysis and control functions. In short, R7FOE is perfectly suited for extreme low-power and energy-harvesting applications.

Today, efficient energy-harvesting technology is facing several challenges. R7FOE eliminates many such challenges. It houses a unique and configurable energy harvest controller (EHC) function, which minimises costly external components. This EHC helps in direct connection to many different types of ambient energy sources such as solar, vibration or piezo-electric. It also manages charging of external power storage devices such as super capacitors or optional rechargeable batteries.

Yoshikazu Yakota, executive vice president and general manager -

Industrial Solution Business Unit, Renesas, says, "I am very pleased that Renesas has achieved this milestone product using SOTB technology, which is first of its kind in the energy-harvesting market."

The SOTB-based embedded controller finds applications in industrial, business, residential, agricultural, healthcare and public infrastructures. It can also be applied to health and fitness apparel, shoes, wearables, smart watches and drones.

Key features of R7FOE embedded controller are:

- CPU: Arm Cortex MO +
- Operating frequency: up to 32MHz and up to 64MHz in boost mode
- Memory: up to 1.5MB flash, 256kB SRAM
- Current consumption while operating at 3.0V
- Active: 20 micron-ampere
- Deep standby: 150 nano-ampere with real-time clock source and reset manager
- Software standby: 400 nano-ampere
- EHC: interface for direct robust connection to energy-generating devices and for charge management of energy-storage devices
- Analogue-to-digital-converter (ADC): fourteen-bit, 32kHz operation frequency, three micron-ampere consumption
- Better graphics capability
- Higher security and encryption

It is time for designers to design and implement energy-harvesting technology-based MCUs. Existing MCU manufacturing processes have not been designed keeping power performance of devices in mind. Machine communication data traffic is short and bursty in nature, and devices are usually battery-powered, which necessitates stringent power management for link control and signalling.

Energy harvesting is well-suited for emerging IoT needs. One of the challenges the IoT brings forth for the network is that devices need to be energy-efficient. Renesas' new energy-harvesting-based MCU, R7FOE, based on SOTB process, can help realise this dream, making IoT-connected devices smarter. **EFY**